

RAG Project Documentation

A comprehensive reference guide containing your project setup requirements, sample knowledge base data, and LangChain document loading troubleshooting steps.

1. Project Requirements

Dependencies for setting up a Retrieval-Augmented Generation (RAG) pipeline using LangChain, OpenAI, Mistral AI, and ChromaDB. Save this block as `requirements.txt`:

```
# Core RAG Framework
langchain
langchain-core
langchain-community

# LLM Providers
openai
langchain-openai
mistralai
langchain-mistralai

# Vector Database
chromadb

# Tokenization & Local Embeddings
tiktoken
sentence-transformers

# Document Loaders & Parsers
pypdf
beautifulsoup4

# Environment Variables Management
python-dotenv
```

2. Sample Knowledge Base Data

Save the following text as `sample_rag_data.txt`. This structured data is designed to test rule-based queries, procedural questions, and entity extraction.

Acme Corp Internal Knowledge Base - Sample Data for RAG Testing

1. Human Resources: Paid Time Off (PTO) Policy

All full-time employees at Acme Corp are entitled to 20 days of Paid Time Off (PTO) per calendar year. Employees can carry over a maximum of 5 days into the next year. Any additional unused PTO will be forfeited. For sick leave, employees are granted 8 days per year, which do not roll over. To request PTO, employees must submit a request through the Workday portal at least two weeks in advance for vacations longer than 3 days.

2. IT Security Guidelines

Password Policy: Passwords must be at least 14 characters long, containing at least one uppercase letter, one lowercase letter, one number, and one special character. Passwords must be changed every 90 days.

Phishing: If you receive a suspicious email, do not click on any links.

Forward the email immediately to phishing-report@acmecorp.local and delete the original message.

Remote Work: When working remotely, all employees must connect to the corporate VPN before accessing any internal systems, source code, or databases.

3. Project Alpha: Q3 Initiative

Project Alpha is Acme Corp's flagship initiative for Q3 2026. The goal of Project Alpha is to migrate our legacy on-premise databases to a cloud-native AWS architecture.

- Project Lead: Sarah Jenkins (VP of Engineering)
- Budget: \$1.2 Million
- Deadline: October 15, 2026

The migration will happen in three phases: Phase 1 (Assessment), Phase 2 (Data Transfer), and Phase 3 (System Cutover). During Phase 3, expected in late September, there will be a scheduled system downtime of approximately 4 hours.

4. Office Facilities

The main headquarters is located at 100 Innovation Drive, San Francisco, CA. The office is open from 7:00 AM to 8:00 PM, Monday through Friday. The company cafeteria is located on the 3rd floor and serves lunch from 11:30 AM to 1:30 PM. A quiet room for meditation and prayer is available on the 2nd floor, Room 204.

3. Handling LangChain Document Loading Errors

If you encounter a `FileNotFoundError` when using `TextLoader`, the safest way to load your documents regardless of where you execute the script from is by resolving absolute paths dynamically:

```
import os
from langchain_community.document_loaders import TextLoader

# Get the absolute directory where this current python script is located
current_dir = os.path.dirname(os.path.abspath(__file__))

# Safely join that folder path with your file name
file_path = os.path.join(current_dir, "sample_rag_data.txt")

data = TextLoader(file_path)
docs = data.load()

print(f"Loaded {len(docs)} document(s) successfully.")
```

Pro-tip: When debugging file errors on Windows, always ensure your file extensions aren't hidden by the File Explorer (e.g. accidentally naming your file `notes.txt.txt`).